

E-commerce Customer Shopping Analysis Dashboard

1. Project Overview

The E-Commerce Customer Shopping Analysis project focuses on analyzing customer purchasing behaviour, sales performance, and subscription trends using Business Intelligence and Data Analytics techniques. The objective of this project is to transform raw customer transaction data into meaningful insights that support data-driven decision-making and business growth.

Through interactive dashboards, the project provides a comprehensive view of customer demographics, revenue patterns, product performance, shipping preferences, and customer segmentation.

2. Problem Statement

Organizations generate large volumes of customer transaction data but often struggle to extract actionable insights. Without proper analysis, businesses may miss opportunities to improve customer retention, increase sales, and optimize marketing strategies.

This project aims to analyze customer and sales data to identify revenue drivers, customer segments, purchasing patterns, and business growth opportunities.

3. Project Objectives

- Analyze customer demographics and purchasing behaviour.
- Identify top-performing product categories and revenue sources.
- Evaluate customer subscription trends and engagement.
- Monitor sales performance across categories, seasons, and shipping methods.
- Generate actionable business recommendations through data visualization and analytics.

4. Tools & Technologies Used

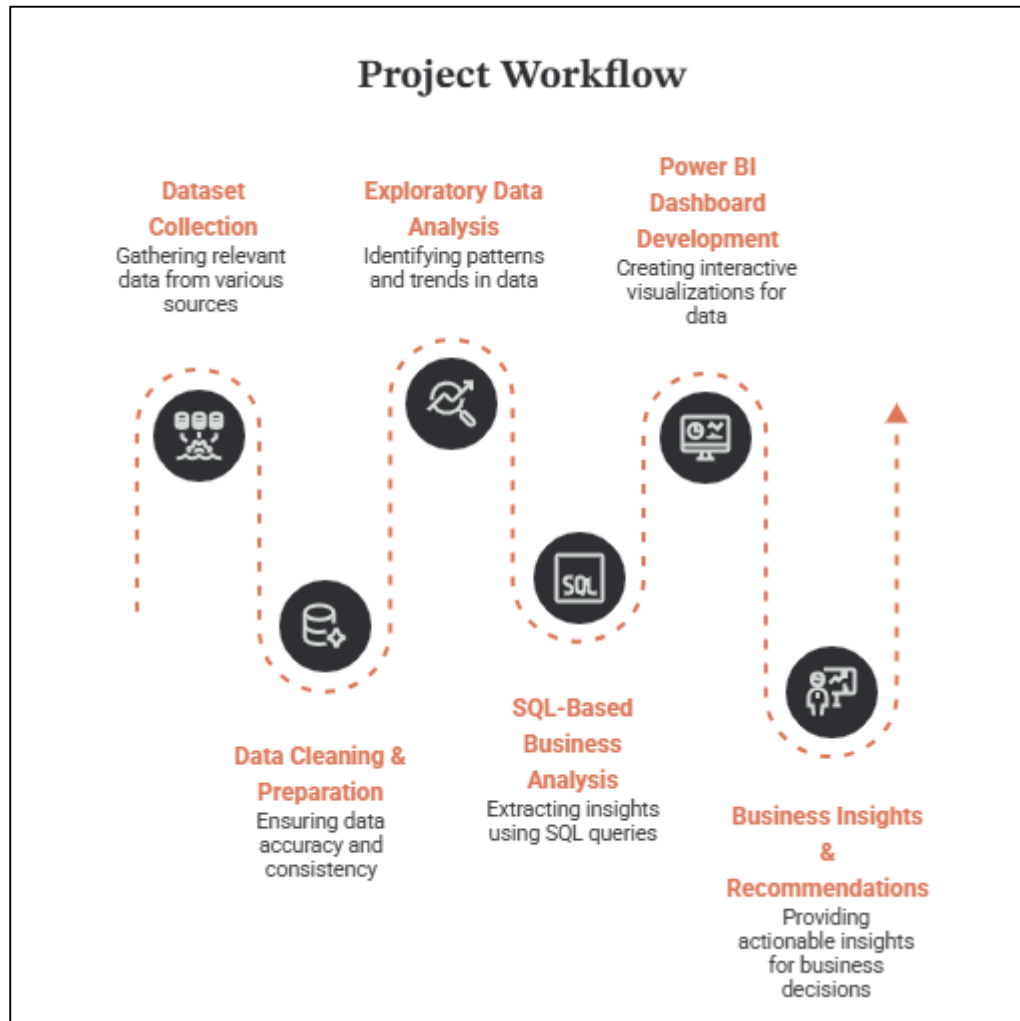
- Power BI
- Python
- SQL
- Excel
- Pandas

- NumPy

5. Dataset Information

- Dataset Name: E-Commerce Customer Shopping Dataset
- Total Records: 3900
- Columns: 18
- Features Included:
 - Customer ID
 - Gender
 - Age
 - Category
 - Purchase Amount
 - Subscription Status
 - Shipping Type
 - Season
 - Review Rating
 - Discount Applied
 - Previous Purchases
- Missing Data: 37 values in Review Rating column.

6. Project Architecture



This project was executed through an end-to-end analytics workflow, starting with data collection, cleaning, and preparation using Python. Exploratory Data Analysis was conducted to understand customer behaviour and sales trends, followed by SQL-based business analysis to answer critical business questions. The insights were then visualized through interactive Power BI dashboards, enabling comprehensive analysis of customer demographics, purchasing patterns, and revenue performance. The workflow concluded with business insights and strategic recommendations aimed at improving customer engagement, sales growth, and overall business performance.

7. Exploratory Data Analysis (Python)

The project began with data preparation, cleaning, and exploratory data analysis using Python to ensure data quality and prepare the dataset for further analysis.

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**

- Created **age_group** column by binning customer ages.
- Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if [discount_applied](#) and [promo_code_used](#) were redundant; dropped [promo_code_used](#).
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

8. Data Analysis using SQL

Performed structured analysis in PostgreSQL to answer key business questions:

- 1) **Customer Segmentation-** Segment customers into New, Returning, and Loyal customers based on their purchase history and determine their revenue contribution.

	customer_segment text	total_customers bigint	total_revenue numeric	avg_spending numeric
1	Loyal Customer	1549	92984.00	60.03
2	Returning Customer	1567	92533.00	59.05
3	New Customer	784	47564.00	60.67

- 2) **Subscription Impact-** Compare average purchase amount, total revenue, and purchase frequency between subscribers and non-subscribers.

	subscription_status text	total_customers bigint	avg_purchase_amount numeric	total_revenue numeric	avg_previous_purchases numeric
1	No	2847	59.87	170436.00	25.08
2	Yes	1053	59.49	62645.00	26.08

- 3) **Revenue by Product Category-** Identify the best-performing categories and their contribution to overall sales.

	category 	total_orders 	total_revenue 
	text	bigint	numeric
1	Clothing	1737	104264.00
2	Accessori...	1240	74200.00
3	Footwear	599	36093.00
4	Outerwear	324	18524.00

- 4) **Revenue by Age Group-** Analyze customer spending across different age groups to identify high-value demographics.

	age_group 	total_customers 	total_revenue 	avg_spending 
	text	bigint	numeric	numeric
1	Young Adult	1028	62143.00	60.45
2	Middle-aged	986	59197.00	60.04
3	Adult	942	55978.00	59.42
4	Senior	944	55763.00	59.07

- 5) **Revenue by Gender-** Compare revenue, average spending, and purchase frequency between male and female customers.

	gender 	total_customers 	total_revenue 	avg_spending 	avg_previous_purchases 
	text	bigint	numeric	numeric	numeric
1	Female	1248	75191.00	60.25	24.60
2	Male	2652	157890.00	59.54	25.71

- 6) **Discount Impact-** Determine whether customers who receive discounts spend more than those who do not.

	discount_applied 	total_orders 	avg_spending 	total_revenue 
	text	bigint	numeric	numeric
1	No	2223	60.13	133670.00
2	Yes	1677	59.28	99411.00

- 7) **Revenue by Shopping Method-** Evaluate customer preferences and the revenue contribution of different shipping options.

	shipping_type text	total_orders bigint	total_revenue numeric	avg_order_value numeric
1	Free Shipping	675	40777.00	60.41
2	Express	646	39067.00	60.48
3	Store Pickup	650	38931.00	59.89
4	Standard	654	38233.00	58.46
5	2-Day Shipping	627	38080.00	60.73
6	Next Day Air	648	37993.00	58.63

8) Seasonal Sales & Revenue- Analyze seasonal trends to identify peak business periods and customer demand patterns.

	season text	total_orders bigint	total_revenue numeric	avg_order_value numeric
1	Fall	975	60018.00	61.56
2	Spring	999	58679.00	58.74
3	Winter	971	58607.00	60.36
4	Summer	955	55777.00	58.41

9) Customer Satisfaction VS Spending- Examine whether customers providing higher review ratings tend to spend more.

	rating numeric	total_customers bigint	avg_spending numeric	total_revenue numeric
1	5.0	68	64.35	4376.00
2	4.9	162	63.75	10327.00
3	4.8	144	61.88	8911.00
4	4.7	148	59.28	8774.00
5	4.6	170	57.31	9743.00

10) Most Profitable Customer Profile- Identify the most profitable customer profile by analyzing age group, gender, category, and spending patterns.

	age_group text	gender text	category text	total_orders bigint	total_revenue numeric
1	Young Adult	Male	Clothing	330	19603.00
2	Senior	Male	Clothing	300	17711.00
3	Middle-aged	Male	Clothing	279	16898.00
4	Adult	Male	Clothing	272	16416.00
5	Young Adult	Male	Accessori...	221	13167.00

9. Data Preparation

The following data preparation steps were performed:

- Data cleaning and validation
- Handling missing values
- Data type conversion
- KPI creation
- Customer segmentation
- Data transformation for reporting

10. Dashboard Structure

1) Home Page

Provides project introduction and dashboard navigation.

2) Executive Overview Dashboard

Key Metrics:

- Total Revenue
- Total Customers
- Average Order Value
- Subscription Rate

Analysis:

- Revenue by Category
- Revenue by Season
- Revenue by Shipping Type
- Subscription Status

3) Customer Insights Dashboard

Analysis:

- Revenue by Age Group
- Sales Count by Age Group
- Revenue Contribution by Gender
- Purchase Frequency by Gender
- Customer Segmentation

4) Product & Sales Analysis Dashboard

Analysis:

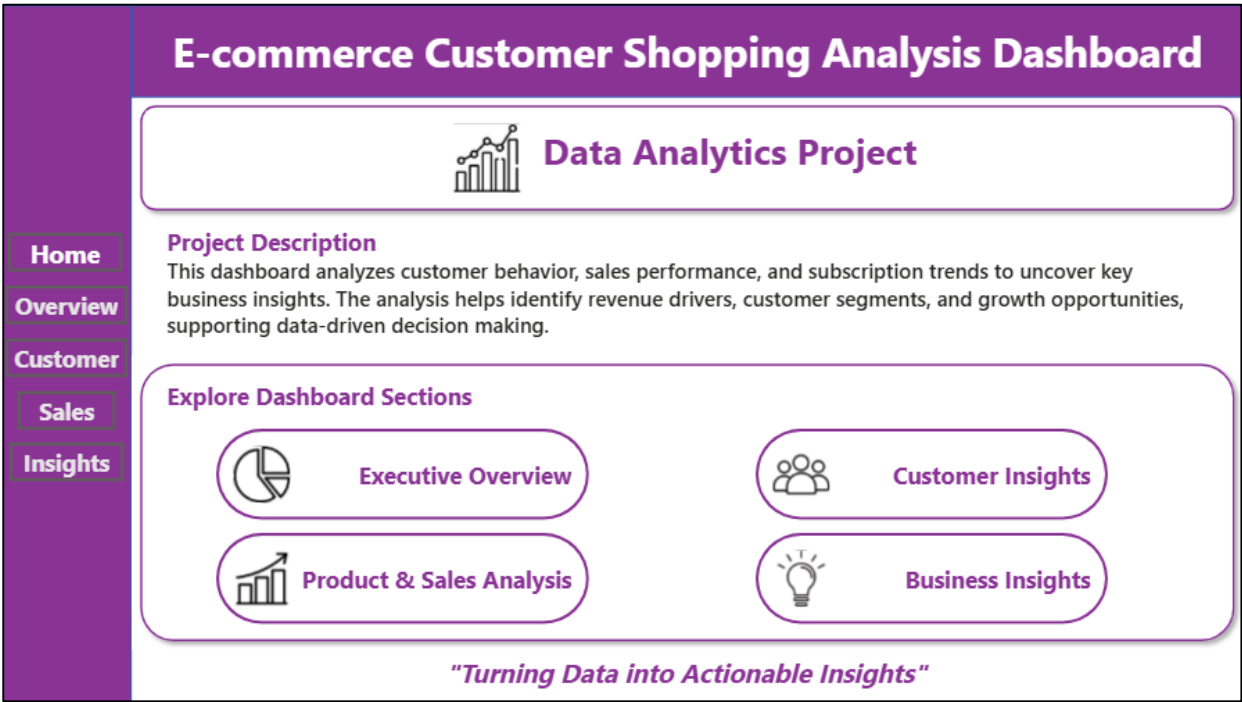
- Revenue by Category
- Sales Count by Category
- Shipping Type vs Revenue
- Shipping Preference
- Revenue by Season
- Purchases by Season
- Discount Impact on Sales

5) Business Insights & Recommendations Dashboard

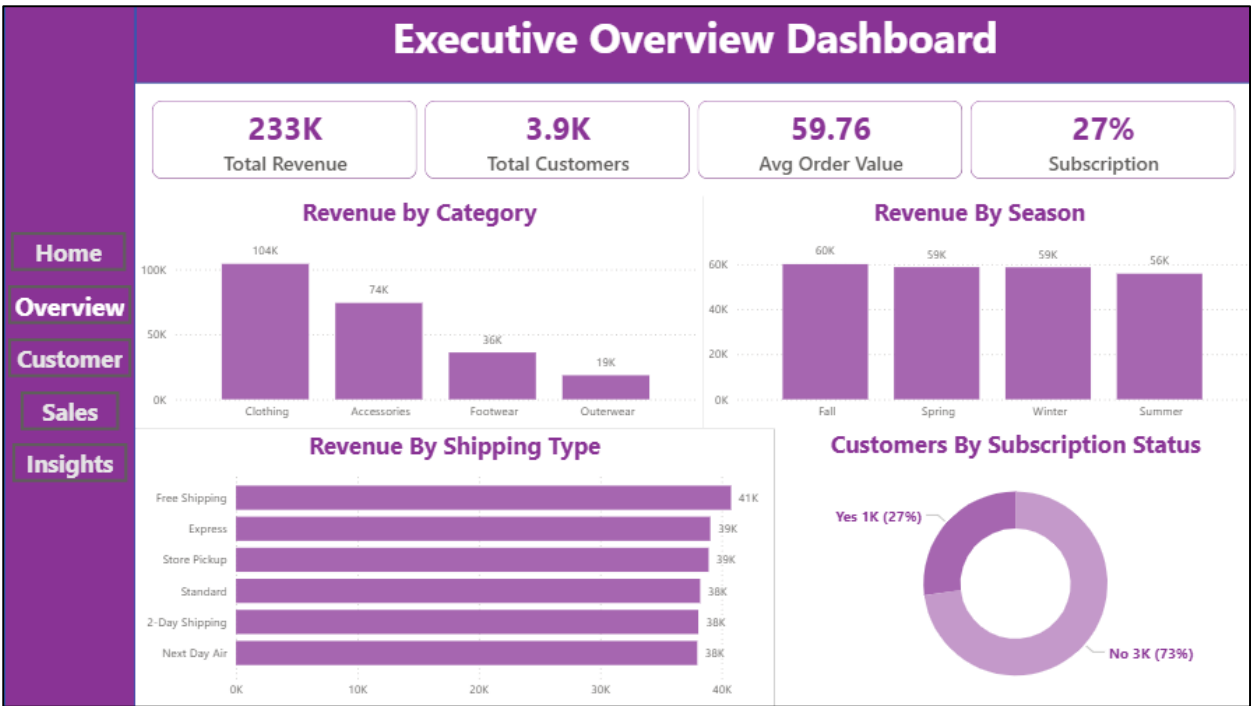
Analysis:

- Key Business Findings
- Strategic Recommendations
- Project Summary
- Conclusion

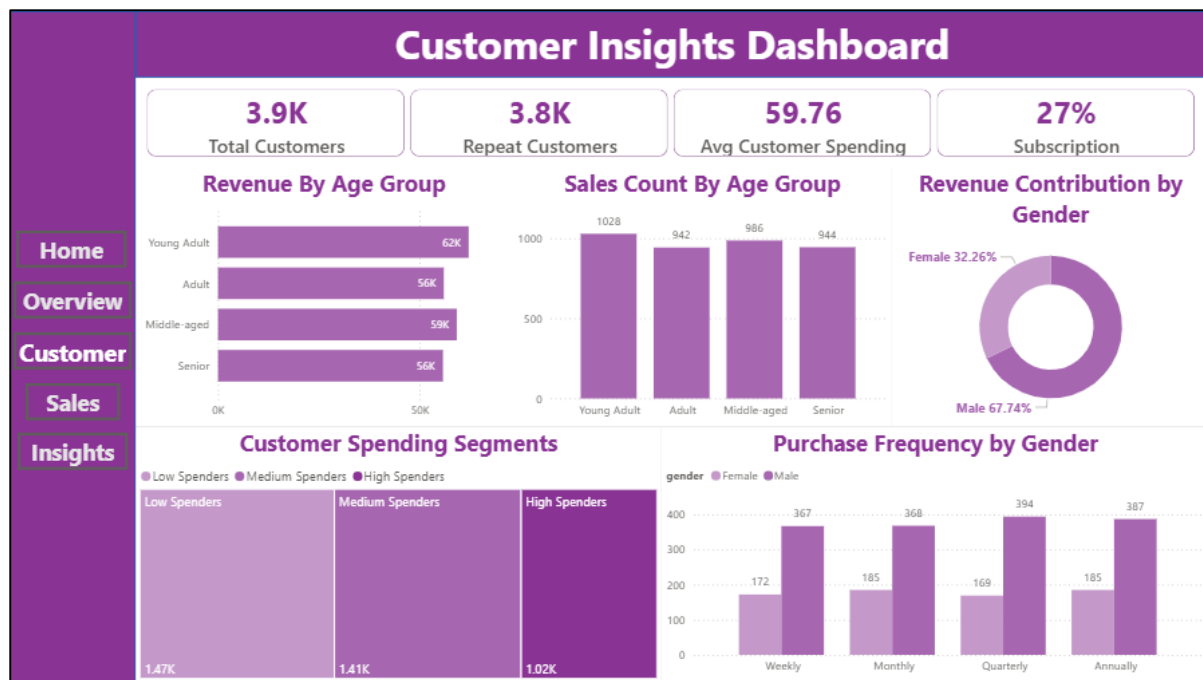
11. Dashboard Screenshots



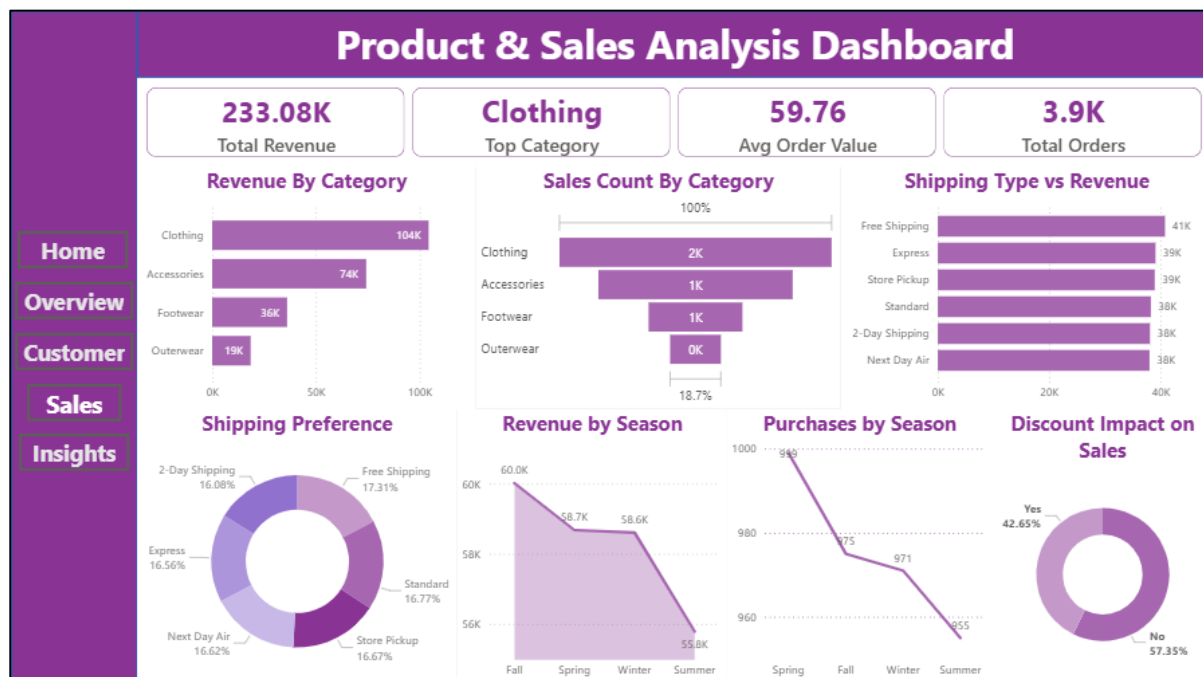
Provides project introduction and navigation to different dashboard sections.



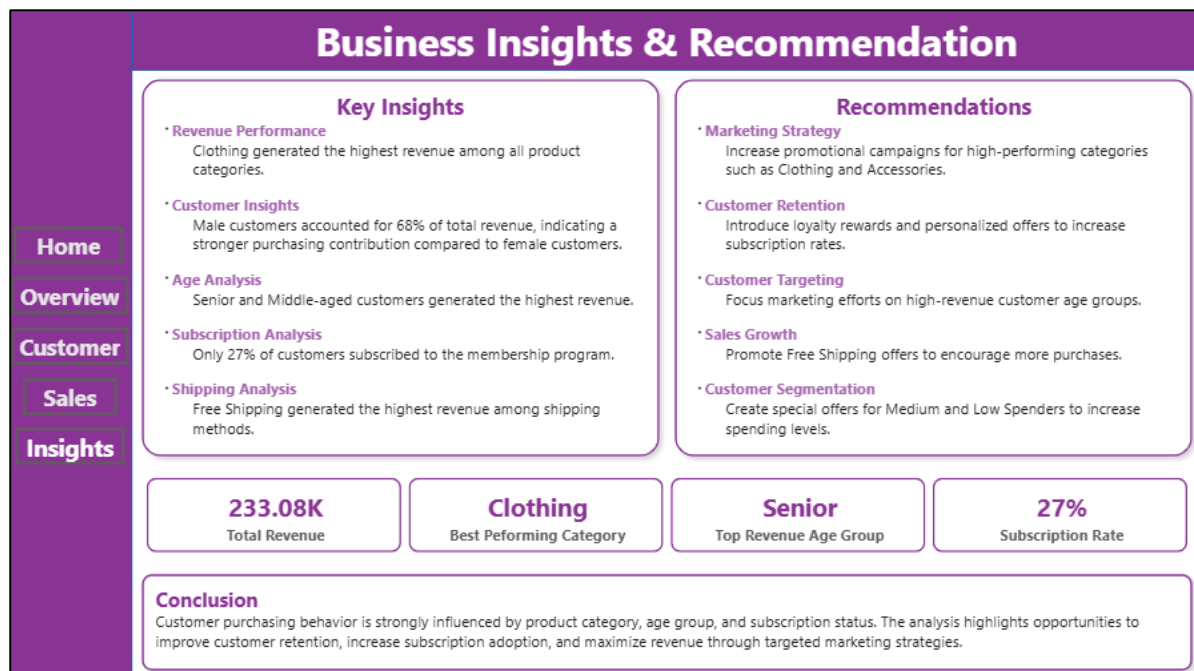
Provides a high-level overview of business performance through key metrics and revenue analysis.



Analyzes customer demographics, spending behaviour, and segmentation.



Evaluates category performance, shipping preferences, seasonal sales, and discount impact.



Summarizes findings and provides strategic business recommendations.

12. Key Performance Indicators (KPIs)

- **Total Revenue-** Measures the total sales revenue generated.
 - Formula: $\text{SUM}(\text{customer}[\text{purchase_amount}])$
- **Total Customers-** Measures the total number of customers.
 - Formula: $\text{DISTINCTCOUNT}(\text{customer}[\text{customer_id}])$
- **Average Order Value-** Measures average spending per transaction.
 - Formula: $\text{Total Revenue} / \text{Total Orders}$
- **Subscription Rate-** Measures the percentage of subscribed customers.
 - Formula: $\text{DIVIDE}(\text{CALCULATE}(\text{COUNTROWS}(\text{customer}), \text{customer}[\text{subscription_status}] = \text{"Yes"}), \text{COUNTROWS}(\text{customer}))$

13. Key Insights

- Clothing generated the highest revenue among all categories.
- Male customers contributed the majority of total revenue (68%).
- Senior and Middle-aged customers generated higher revenue compared to other age groups.
- Free Shipping contributed the highest revenue among shipping methods.
- Subscription adoption remained relatively low (Only 27%).

14. Recommendations

- Promote high-performing product categories.
- Increase subscription adoption through targeted campaigns.
- Focus marketing efforts on high-value customer segments.
- Implement loyalty and retention programs.
- Develop personalized offers for medium and low spenders.

15. Project Outcome

The project successfully transformed raw customer transaction data into interactive business intelligence dashboards. Through data cleaning, SQL analysis, KPI development, and visualization, the project identified customer purchasing patterns, revenue drivers, and business growth opportunities, enabling data-driven decision-making.

16. Conclusion

The E-Commerce Customer Shopping Analysis project demonstrates how data analytics can be used to understand customer behaviour, improve business performance, and support strategic decision-making. By leveraging interactive dashboards and business intelligence techniques, organizations can gain valuable insights and make informed decisions to enhance customer satisfaction and profitability.